07-20 第1篇

阅读理解（说明文） How Do Our Ears Hear Sound?

① Close your eyes for ten seconds and listen. You may hear a clock on the wall, a bird outside, a bus in the distance, or a classmate turning a page. The world is full of quiet and loud sounds — and our ears, small as they are, can catch almost all of them. But have you ever stopped to ask: **how exactly does an ear turn a sound from the air into a thought in our brain?**

② Every sound begins with a **vibration**. When someone speaks, the vocal cords shake quickly. When a guitar string is pulled, it moves back and forth. These movements push the air around them. The air pushed first pushes the next bit of air, and so on. The result is an invisible wave that travels through the air — we call it a **sound wave**. Sound waves can also travel through water and through hard things, like a desk or a metal pipe. They cannot, however, travel in empty space. That is why astronauts on the Moon must use radios even when they stand face to face.

③ The ear has three main parts: the **outer ear**, the **middle ear**, and the **inner ear**. The outer ear is the curvy shape on the side of our head. It works a bit like a little satellite dish: it catches the sound wave from the air and sends it down a short tunnel — the ear canal — to a thin, stretched skin called the **eardrum**. When the wave hits the eardrum, the eardrum moves, just as a drum moves when a child hits it with a stick.

④ The middle ear is a small room behind the eardrum. Inside it are **three tiny bones**, the smallest bones in the human body. They have interesting names: hammer, anvil and stirrup. When the eardrum shakes, these three bones pass the movement along, like passing a ball from hand to hand. Each bone makes the movement a little stronger, so that even a very soft sound becomes clear enough for the inner ear to read.

⑤ The inner ear is where the real magic happens. It has a curled, snail-shaped tube filled with liquid, called the **cochlea**. Inside the cochlea are thousands of tiny hair-like cells. When the small bones push the liquid, the liquid moves, and the hair cells wave back and forth. Each hair cell is good at catching a different kind of vibration — some listen for low sounds (a big drum), some for high sounds (a bird song). The hair cells turn the movement into **electrical signals** and send them along a nerve to the brain. The brain then does something amazing: in less than a quarter of a second, it decides whether we are listening to Mum's voice, a phone, or the wind.

⑥ Our ears do more than listen. They also help us keep our **balance**. Next to the cochlea there are three tiny loops full of liquid. When we turn our head, the liquid moves, and the

brain quickly understands which way we are facing. This is why riding a fast ride at the park can make us feel dizzy — the loops cannot catch up with the sudden changes.

⑦ Because the inner ear is so delicate, we should take care of it. Loud earphones for long hours, swimming in very cold dirty water, or putting sharp things into the ear can damage the tiny hair cells. Once these cells are lost, they do not grow back. So turn the volume down, take short listening breaks, and remember: behind every song you love, there are thousands of little hair cells quietly doing their job.

根据文章内容，选择最佳答案。

1. According to paragraph ②, why do astronauts on the Moon use radios even when they are close to each other?

A. Because sound waves cannot travel in empty space.
B. Because the Moon is too cold for talking.
C. Because radios are cheaper than talking.
D. Because they cannot open their mouths on the Moon.

2. What is the main job of the eardrum?

A. To store food for the body.
B. To change shape when a sound wave hits it, so the movement can pass into the middle ear.
C. To produce sound for the outer world.
D. To keep the ear warm in winter.

3. What do the three tiny bones in the middle ear do?

A. They protect the brain from cold water.
B. They pass the movement from the eardrum along, making the sound clear enough for the inner ear.
C. They store memories of songs.
D. They create sound waves all by themselves.

4. According to the passage, what can damage the tiny hair cells in the inner ear?

A. Listening to soft music at a low volume.
B. Keeping the ears clean with a soft towel.
C. Loud earphones used for long hours, dirty water, or sharp things put into the ear.
D. Looking at the sky for too long.

5. What is the main purpose of this passage?

A. To sell a new kind of earphone.
B. To explain how the ear turns sound waves into signals the brain can understand, step

by step.
C. To tell a story about a boy who could not hear.
D. To compare the ears of humans with those of cats.

答案与解析

1.
【答案】A
【题型】细节理解题
【定位】第②段 "They cannot, however, travel in empty space..."
【解析】声波不能在真空传播，故需无线电。B/C/D 荒谬。故选 A。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "When the wave hits the eardrum, the eardrum moves..."
【解析】鼓膜振动以传递声波。A/C/D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第④段 "pass the movement along... makes the movement a little stronger..."
【解析】三块小骨头传递并放大振动。A/C/D 均错。故选 B。

4.
【答案】C
【题型】细节理解题
【定位】第⑦段 "Loud earphones... swimming in very cold dirty water, or putting sharp things..."
【解析】大音量、脏水、尖物均会损伤毛细胞。A/B 有益；D 荒谬。故选 C。

5.
【答案】B
【题型】主旨大意题
【定位】通篇
【解析】全文逐步解释耳朵如何把声波转化为大脑信号。A/C/D 均非主旨。故选 B。

词汇注释（超纲词）

- **vibration** [n.] 振动
- **eardrum** [n.] 鼓膜
- **cochlea** [n.] 耳蜗
- **anvil / stirrup** [n.] 砧骨 / 镫骨（中耳小骨）

- **delicate** [adj.] 精密的、易损的